



Skin Disease Classification with Fairness Auditing

An AI-based web tool for image-based skin disease detection

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Problem Statement

Prevalence

Skin diseases, are among the most common health conditions worldwide

Access Gap

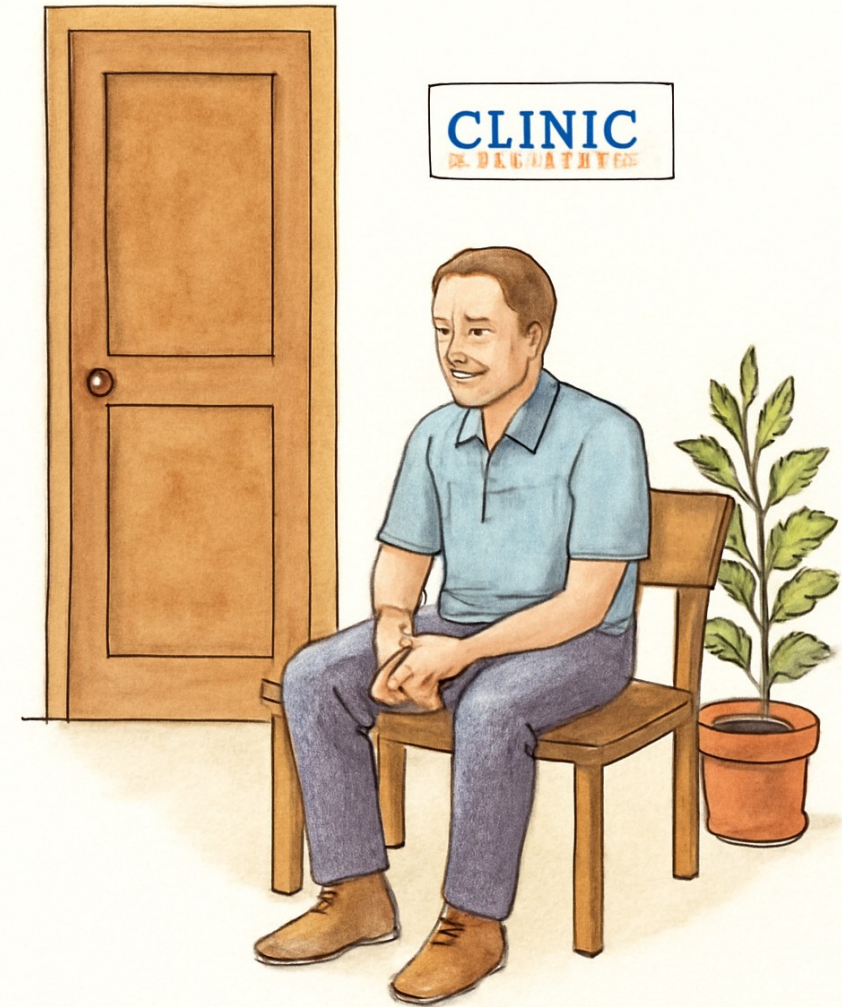
Limited access to dermatologists, especially in rural and underserved areas

Delayed Diagnosis

Worse prognosis and higher treatment costs when detection is late

Early Detection

Significantly improves treatment outcomes when caught early



Motivation & Objectives

Motivation

- AI can bridge the gap between patients and specialist dermatological care
- Enable fast, accessible preliminary screening at scale

Objectives

- 1 Build a CNN-based classification model
For common skin lesion types
- 2 Evaluate fairness across skin tones
Fitzpatrick scale analysis
- 3 Deploy as a web application
User-friendly image classification interface

Dataset: HAM10000



Source

Harvard Dataverse / Kaggle

Size

~10,015 dermatoscopic images

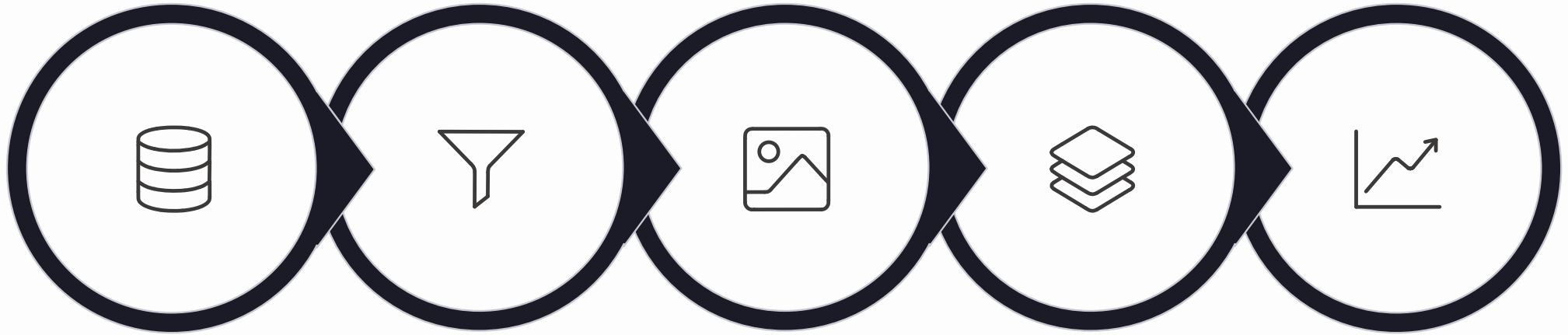
7 Categories

Melanoma · Melanocytic Nevus ·
Basal Cell Carcinoma · Actinic
Keratosis · Benign Keratosis ·
Dermatofibroma · Vascular Lesion

Known Challenge

Class imbalance — melanocytic
nevi dominate the dataset

Proposed Methodology



Data
Collection

Preprocessin
g

Data
Augmentatio
n

Transfer
Learning

Fine-Tuning &
Eval

Class Imbalance

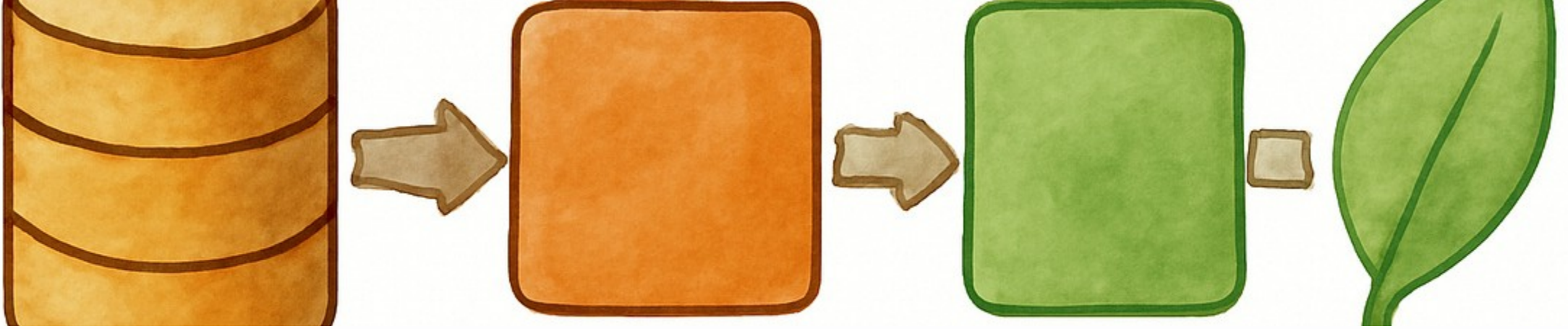
Weighted loss functions or
oversampling techniques

Transfer Learning

Pretrained CNN weights for faster
training and better generalization

Data Augmentation

Rotation, flipping, brightness
adjustment



Project Execution Workflow

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Phase 1: Data & Baseline Model

Image Handling & Augmentation

- PIL/Pillow and OpenCV: resizing, normalizing, reading image files programmatically
- Data augmentation: rotation, flipping, zoom, color jitter using torchvision.transforms or albumentations
- Critical for handling HAM10000 class imbalance

CNN Fundamentals & Transfer Learning

- Understand why CNNs work better than dense networks for images (convolution and pooling)
- Use pretrained models (EfficientNet or ResNet) from ImageNet
- Fine-tune on HAM10000 instead of training from scratch
- Framework choice: PyTorch (recommended for Grad-CAM support) or Keras/TensorFlow

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Phase 2: Fine-tuning & Fairness Auditing

Evaluation Metrics

- Confusion matrix, precision/recall/F1 per class
- Macro vs. weighted averages (accuracy alone is misleading on imbalanced data)
- Weighted loss functions to handle class imbalance

Fairness & Interpretability

- Grad-CAM: visualize which image regions the model focuses on
- Disaggregate metrics by Fitzpatrick skin-tone group (pandas groupby)
- Present fairness gaps clearly to demonstrate equitable performance

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Phase 3 & 4: Deployment & Validation

- Integrate trained model into Flask/FastAPI backend
- Build web interface for image upload and result display
- Real-world testing and clinic partnerships

Fairness Auditing Component



Why It Matters

- Dermatology datasets historically underrepresent darker skin tones
- Models trained on biased data perform unevenly across patient populations
- Real-world reliability requires evaluation across diverse groups

Our Approach



Fitzpatrick Annotations

Measure accuracy separately per skin tone group



Grad-CAM

Inspect model focus regions during predictions



Fairness Metrics

Equitable performance, not just overall accuracy

System Architecture: Web Application Workflow



Frontend

HTML / CSS / JavaScript — image
upload interface

Backend

Flask or FastAPI — REST API layer

ML Model

Trained CNN via TensorFlow / PyTorch

Real-World Usefulness & Impact



Remote Patients

Instant preliminary screening
without specialist access



Healthcare Systems

Triage tool to prioritize urgent
cases such as suspected
melanoma



Telemedicine

AI-assisted second opinion for
remote consultations



Public Health

Raises awareness and
encourages early detection
across all skin types

📄 Fairness auditing ensures the tool is reliable across **all** populations — not just majority groups.

Tools, Technologies & Timeline

Technology Stack



Languages & Frameworks

Python · TensorFlow / PyTorch · OpenCV



Web Deployment

Flask or Streamlit · HTML / CSS / JavaScript



Explainability

Grad-CAM for model interpretability

Project Timeline

